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De Facto Opioids: Characterization of Novel 7-Hydroxymitragynine and Mitragynine Pseudoindoxyl Product Marketing

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Abstract

Background: Within the past year, the online sale of semi-synthetic products containing chemicals derived from mitragynine, kratom's primary alkaloid: 7-hydroxymitragynine (7-OH) and mitragynine pseudoindoxyl (MP) has emerged. Both are highly selective mu opioid receptor agonists undetectable or not present in fresh kratom leaves and may, as constituents of novel product formulations, pose public health risks.

Method: Between September 2024-February 2025 we abstracted from websites of vendors selling 7-OH and MP products information about these products, including formulation types, dose/serving size, cost, and effect, health, and drug claims

Results: We identified 304 7-OH and MP semi-synthetic products. Most (82.2%) were 7-OH-only products formulated as chewable/sublingual tablets, shots, or gummies; 14.5% were combination 7-OH-MP, and 3.3% MP-only. MP and 7-OH-MP combination products were almost uniformly marketed as "kratom" while 92.0% of 7-OH-only products were advertised as such. Across products' online marketing content, 73.3% made effect claims, most often promising increased focus and/or providing relaxation. Functional claims of pain and anxiety relief were made by 37.8% of products while 12.5% of made drug claims. The mean cost per recommended

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Declaration of Interest Statement

All authors have no competing interests to declare.

dose/serving was \$3.97; MP products had a mean cost closer to \$5 per recommended dose/serving. Several products had names alluding to prescription opioids.

Conclusion: The increasing proliferation and availability of 7-OH and MP products is concerning as these products, and their chemical constituents, have never been studied in humans. Such formulations are not comparable to kratom leaf or extract products despite marketing material suggesting otherwise. Pending further investigation regulatory action should be undertaken.

Keywords

7-hydroxymitragynine; 7-OH, mitragynine pseudoindoxyl; kratom; opioid

1.0 Background

Kratom, a psychoactive botanical taken for centuries in Southeast Asia, is increasingly used in the West for its mild stimulatory and euphoric effects (Cinosi et al. 2015; Smith et al. 2023; Smith, Panlilio, Feldman, et al. 2024). Mitragynine, kratom's most abundant alkaloid that acts at adrenergic, opioid, dopamine, and serotonin receptors, may be responsible for many of kratom's neuropsychiatric effects. (León et al. 2021; Obeng et al. 2022; Ponglux et al. 1994). The chemical 7-hydroxymitragynine (7-OH) is not present in fresh kratom leaves but is an artifact of the drying process (Zhang et al. 2022, Smith, et al. 2024; Sharma, 2025); it is also formed by oxidation of mitragynine by hepatic and intestinal cytochromes. The binding affinity of 7-OH at the mu-opioid receptor (MOR) is 10-22 greater than morphine; 7-OH is further metabolized in humans to mitragynine pseudoindoxyl (MP), a chemical nearly as potent as fentanyl (Chakraborty et al. 2021; Kamble et al. 2020; Kruegel et al. 2016; 2019; Qu et al. 2023). Because 7-OH and MP are not found in fresh, native kratom leaf (Zaremba et al., 1974) products containing these ingredients manufactured through conversion of mitragynine can be considered semi-synthetic derivatives of mitragynine. The safety of 7-OH and MP in humans is unstudied.

Kratom products sold in the West include extracts and isolates, which differ from traditional decoctions made from boiling fresh leaves, a practice that limits the total alkaloids ingested per serving (Brown, Lund, and Murch 2017; Grundmann et al. 2023; Singh, Narayanan, and Vicknasingam 2016). Increasing global demand for kratom has coincided with growing diversity of kratom product formulations that include organic extracts, gummies, vapes, liquid shots, or tablets (Smith, Panlilio, Sharma, et al. 2024). In 2023-2024, however, novel semi-synthetic 7-OH and MP products appeared online with the monikers "7-OH," "7-hydroxy," or "7" and, in the case of MP, as "pseudo." These semisynthetic kratom alkaloids are overwhelmingly mislabeled or marketed as "kratom" even though the chemical composition is wholly distinct (Sharma 2025; Smith et al. 2025).

The emergence of high potency, non-prescription commercial products derived from kratom impels a better understanding of this semi-synthetic market. Therefore, we investigated the online prevalence of 7-OH and MP products; characterized products' dosing, cost, and formulation; and reviewed marketing claims including references to "kratom."

2.0 METHODS

2.1 Data Collection

Between September 2024-February 2025, we monitored online 7-OH and MP marketing content to compile a product database. We identified vendor websites and products through browser searches using keywords “7-hydroxymitragynine,” “7-hydroxy,” “7-OH,” “Mitragynine pseudoindoxyl,” “pseudoindoxyl,” and “pseudo.” We counted products sold in varying sizes only once to avoid duplications in the database. Alternatively, some websites sold similar products (e.g., drink mix) under different flavors. If these contained slightly different formulations and/or made separate claims, we considered them distinct.

2.2 Product and Marketing Content

2.2.1 Product categorization was coded as: “7-OH,” “MP,” or “combination 7-OH and MP.”

2.2.2 Product information was obtained by scraping websites for: URL, brand name, product name, complete amount (mg) in full package, mg in a single dosing unit, mg in recommended dose/serving, online product description, product images, product cost per package, cost per serving, servings per package, customer reviews, and semi-synthetic substance(s).

2.2.3 Product type was classified: chewable/sublingual tablets, shots, gummies, drink mixes, vape pens, syrups, capsules, food products, sublingual strips, or powders.

2.2.3 Product claims were roughly classified: “general wellbeing claims,” “functional claims,” or “drug claims” and identified by reading product descriptions across websites and product packaging, including images when available. “*General wellbeing claims*” were statements linking the semi-synthetic product to beneficial general effects; products were categorized into the following emergent general wellbeing claims: boosting energy, increasing focus, improving wellbeing, improving mood, providing relaxation, and workout supplement. General wellbeing claims that we identified were similar, but not always equivalent, to those of a structure/function claim of a dietary supplement as defined by the U.S. Food and Drug Administration (Borchers, Keen, & Gershwin, 2016). “*Functional claims*” were statements that linked the product to more specific therapeutic-like outcomes; products were categorized into the emergent functional claims: pain/discomfort relief or management, and stress/anxiety reduction. “*Drug claims*” were statements indicating the product conferred some specific therapeutic outcome, such as stimulatory or opioid activity. Note that this claim is not isomorphic with disease claims, which dietary supplements are not permitted to make. We want to emphasize that our categorizations for product claims are not rigid and that, depending on specific marketing language, could correspond to one or more products claims. As such, our three types of “claims” provide only a crude method for categorizing the statements made online for novel 7-OH and MP marketing.

2.3 Data Analysis

Quantitative analyses were done using Excel with findings summarized descriptively. We abstracted and present illustrative quotes but have removed brand names. These appear in the full Supplement 1 database; we strongly encourage readers to view this material.

3.0 RESULTS

We identified 304 products: 82.2% sold as 7-OH, 14.5% as combination 7-OH-MP, and 3.3% as MP. Note that text segments quoted in the results may exemplify more than one product claim type (each type was coded separately and quantified).

3.1 Product Type, Dosage, Claims, and Reviews

Table 1 shows marketed semi-synthetic products by type. Chewable/sublingual tablets were the most common (60.2%), followed by shots (20.7%). Websites stressed the ease of using semi-synthetics compared to kratom. For instance, a 7-OH chewable tablet is marketed as follows:

“The chewable format is designed for convenience, allowing you to simply chew a tablet and savor the berry flavor. This eliminates the need for mixing or preparation... The compact size of each tablet makes them portable and perfect for on-the-go use, so you can carry them in your bag or pocket and take them whenever needed.”

In this way, many websites marketed semi-synthetics as consumer-friendly formulations while highlighting that kratom products required effort to prepare.

Table 1 summarizes products' characteristics and websites and/or packaging that made specific claims or misleadingly presented their product as kratom. Most (73.4%) of products made what we considered to be *general wellbeing claims*, the most common being increased focus (58.2%), provides relaxation (46.7%), or boosts energy (39.5%). Products sold as 7-OH had a greater proportion of effect claims (76.8%) than MP (50.0%) or combination products (59.1%). Emblematic of the effect claims is one 7-OH vendor's description of the benefits customers might expect:

“Enhanced Relaxation, Focus, Energy Motivation, Mood Elevation, Body Buzz, Euphoria, Discomfort Suppression, Apprehension Relief, Feeling Present, Stimulation, Arousal, Social Lubrication”

Functional claims were made by 37.8% of labels or websites, generally related to pain relief (26.3%) stress/anxiety reduction (21.4%) effects. One 7-OH product touted:

“Each tablet contains 15mg of 7-Hydroxymitragynine, a compound renowned for its strong analgesic and mood-enhancing properties. This makes ‘[Brand] Kratom Chewable Tablets’ an ideal choice for those seeking quick relief from pain, anxiety, or stress, as well as those looking to improve their overall sense of well-being.”

Some branded their products to imply curative powers associated with prescriptions. For example, the brand names included terms such as “7rx,” “Curevana,” “Generic RX,”

“Pain Crusher Rx.” Many products were packaged in ways that mimicked FDA-approved medications (e.g., cough syrup, blister packs).

Few vendor websites and/or packaging made *drug claims* about their semi-synthetic product (12.5%). Most included content related to opioid pharmacology. For instance, one MP tablet product description explained to consumers that:

“Pseudoindoxyl, the active compound in these tablets, is known for its potent analgesic properties, making it an excellent choice for those dealing with chronic pain or discomfort. It works by interacting with the body’s opioid receptors, providing significant relief without the typical side effects associated with traditional opioids.”

Another drug claim (which was interspersed with other claims) that we identified included the potential for sedation, wherein a 7-OH tablet vendor stated:

“[A]t higher doses, 7-Hydroxymitragynine may induce sedation, offering a calming effect that can be particularly helpful for individuals suffering from insomnia. This sedative quality can promote better sleep and reduce tension, contributing to overall well-being.”

Other products were also promoted for stimulant effects; one 7-OH chewable product was described as:

“Unlike some stimulants, 7-OH delivers a smooth, sustained energy boost without the jitteriness or crash, ensuring a balanced and effective experience.”

3.2 Misleading Marketing

Most products were misleadingly marketed as kratom or *Mitragyna speciosa* (93.1%). Mislabeling occurred in naming (e.g., ‘Kratom 7-hydroxymitragynine Extract Chew’), packaging (e.g., ‘Pure Kratom Alkaloid’), and/or in the descriptions (e.g., ‘[this shot] provides a powerful, convenient, and fast-acting way to enjoy the benefits of kratom’). One entitled “Benefits of 7-OH” elaborated:

“[K]ratom's mood-enhancing qualities can alleviate stress and anxiety, fostering a sense of calm and positivity. Many find relief from discomfort through kratom's analgesic properties, which may help manage chronic pain conditions effectively... its relaxing effects can aid in achieving restful sleep, contributing to overall mental and physical rejuvenation. Kratom's versatility extends to its use as a tool for managing withdrawal symptoms from other substances, providing support during recovery processes. As a natural botanical, kratom offers a holistic approach to wellness, appealing to those seeking alternatives to synthetic medications.”

3.3 Dosing and cost

Products contained a wide range (1-700 mg) of 7-OH and/or MP in a single dose/serving and per recommended dose/serving (Table 2). The average recommended dose/serving was 20mg. Compared to the other semi-synthetics, MP products had the lowest mean

recommended dose (10.1mg). Irrespective of product, marketing focused on ease of dosing. One 7-OH vendor stated:

“Every pouch includes four expertly pressed classic extract tablets. These tablets are scored down the center for easy dosing and contain two servings total. Each serving is 7.5mg of pure 7-OH extract, for 15mg total per tablet.”

The mean cost per recommended dose was \$3.97; MP products had a mean cost closer to \$5 per recommended dose. Lastly, while some products had many customer reviews (up to 143 for 7-OH), most had zero or few reviews.

4.0 DISCUSSION

We identified widespread online marketing of semi-synthetic products containing ingredients which, if accurate, have high MOR binding affinity. We identified that 7-OH as the most common constituent of commercial semi-synthetic products. The difficulty and expense of isolating 7-OH from kratom leaves, but also the sheer amount of active ingredient, strongly suggests that these semi-synthetic products are manufactured in a laboratory from purified mitragynine. The chemical processes used to make 7-OH also likely places consumers at risk for inadvertent exposure to toxicity from synthetic reagents and reaction by-products. Novel 7-OH products, therefore, carry fundamentally different risks than those presented by the kratom market that existed in the US for nearly 20 years (Boyer et al. 2008; Smith et al. 2022).

The trend toward increasingly potent semi-synthetic chemicals based upon kratom alkaloids is concerning. Product descriptions frequently hinted at neuropsychiatric opioidergic effects with terms such as “buzz” or “euphoria” coupled with brand names that evoke opioids (e.g., Roxy, Opia, Hydroxie, Perks). Consumers may not appreciate that 7-OH products may induce tolerance and withdrawal symptoms if used regularly, and can, for some, include risks for developing addiction.

In some respects, the widespread availability of these semi-synthetic alkaloids is reminiscent of internet ‘pharmacies’ that sold oxycodone and other opioids in the 2000s without prescription. Just as the online pharmacies once did, online 7-OH vendors may use discreet packaging and direct-to-home shipping practices to obviate state laws intended to protect consumers. Whereas kratom products have been adulterated with hydrocodone and morphine (LeSaint et al., 2022), the potency of semi-synthetic kratom alkaloids may obviate the need for adulteration with opioids.

Marketing of these products poses problems. First, labeling 7-OH and MP products as “kratom” misleads consumers who intend to use a low-potency botanical supplement but instead ingest a high-potency opioid. Second, websites may persuade consumers that 7-OH and MP formulations are chemically pure and consistently manufactured; however, online photographs from some manufacturers themselves show several vials of powdered 7-OH, each a different shade in color that indicates differences in composition and purity. Inconsistencies in manufacturing and non-adherence to good manufacturing practices likely lead to variation in dose and composition of products, which, due to the

MOR binding potency of 7-OH and MG, affects overdose risk and toxicity. Finally, the marketing material for some formulations (e.g., sublingual tablets) ignore their products' increased bioavailability which, when coupled with relatively shorter duration of action of 7-OH, heightens the risk of dependence or addiction (Maxwell et al. 2021; Sharma 2025). We conclude that the widespread availability of these semi-synthetics represents an unrecognized but growing public health threat

Our methodology has limitations. Products sold in brick-and-mortar stores were not included in this sample, although the location of sales would not affect the pharmacology under discussion (Hemby et al. 2019). We did not chemically analyze products or interview consumers; this work, however, is underway and beyond the scope of this exploratory project. We are confident that health risks of 7-OH and MG are real, as forensic toxicologists have identified 7-OH in postmortem samples (Osawa and Johnson 2025). Given the pace with which these products are emerging, further investigation and clinical awareness is warranted.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- Synthetic 7-hydroxymitragynine products are misleadingly marketed as kratom
- Synthetic mitragynine pseudoindoxyl products are misleadingly marketed as kratom
- Marketed products have high mu opioid receptor affinity and no human safety data
- Drug and health claims do not conform to supplement regulation but are legally sold
- Cost per recommended dose across these novel products averaged \$3.97

Table 1.

Characteristics of Products Sold as 7-Hydroxymitragynine (7-OH), Mitragynine pseudoindoxyl (MP), or Both

Characteristics	All Products (N=304)		Products Sold as 7-OH (n=250)		Products Sold as MP (n=10)		Products Sold as Both 7-OH and MP (n=44)	
	Freq	%	Freq	%	Freq	%	Freq	%
Product Type								
Chewable or Sublingual Tablet	183	60.20	140	56.00	5	50.00	38	86.36
Shot	63	20.72	55	22.00	4	40.00	4	9.09
Gummy	13	4.28	13	5.20	0	0.00	0	0.00
Drink Mix	12	3.95	12	4.80	0	0.00	0	0.00
Vape Pen	9	2.96	9	3.60	0	0.00	0	0.00
Syrup	7	2.30	6	2.40	1	10.00	0	0.00
Capsule	6	1.97	6	2.40	0	0.00	0	0.00
Sublingual Strip	6	1.97	6	2.40	0	0.00	0	0.00
Food Product	4	1.32	2	0.80	0	0.00	2	4.55
Powder	1	0.33	1	0.40	0	0.00	0	0.00
Website / Packaging Product Claims								
General Wellbeing Claims	223	73.36	192	76.80	5	50.00	26	59.09
Claims to Provide Increased Focus	177	58.22	152	60.80	2	20.00	23	52.27
Claims to Provide Relaxation	142	46.71	125	50.00	3	30.00	14	31.82
Claims to Boost Energy	120	39.47	101	40.40	2	20.00	17	38.64
Claims to Improve Wellbeing	114	37.50	97	38.80	2	20.00	15	34.09
Claims to Improve Mood	110	36.18	97	38.80	2	20.00	11	25.00
Claims to be a Workout Supplement	8	2.63	8	3.20	0	0.00	0	0.00
Functional Claims	115	37.83	100	40.00	3	30.00	12	27.27
Claims to Pain/Discomfort Relief	80	26.32	70	28.00	2	20.00	8	18.18
Claims to Reduce Anxiety/Stress	65	21.38	54	21.60	2	20.00	9	20.45
Drug Claims	38	12.50	33	13.20	2	20.00	3	6.82
Marketed/Defined as Kratom	283	93.09	230	92.00	10	100.00	43	97.73

Table 2.

Dosage, Cost, and Number of Reviews for Products Sold as 7-Hydroxymitragynine (7-OH), mitragynine pseudoindoxyl (MP), or Both

Characteristics	All Products (N=304)					Products Sold as 7- OH (n=250)					Products Sold as MP (n=10)					7-OH (n=10)	
	Min	Max	Mdn	Mean	SD	Min	Max	Mdn	Mean	SD	Min	Max	Mdn	Mean	SD	Min	Max
Total Package (mg)	2	7000	60	122	423	2	7000	58.00	120.48	465.03	15	240	50.00	70.20	64.94	30	400
Single Dose/ Serving (mg)	2	750	20	35	72	2	750	18.00	36.41	78.97	4	40	15.00	16.52	9.73	10	100
Recommended Dose/Serving (mg)	1	700	10	20	57	1	700	10.00	21.92	63.01	4	20	9.50	10.12	4.42	9	50
Cost of Package (\$)	3.95	139.95	19.49	22.86	14.78	3.95	139.95	18.47	21.97	14.90	11.49	90.00	16.29	27.46	25.42	6.50	64.95
Cost Per Recommended Dose/Serving (\$)	1.14	17.99	3.37	3.97	2.36	1.14	17.99	3.37	4.04	2.52	2.44	8.25	4.50	4.90	2.15	1.94	6.24
Number of Product Reviews	0	143	0	3	12	0	143	0.00	3.45	13.12	0	9	0.50	1.90	3.11	0	25